Model and Variable Selection

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Intended Learning Outcomes

- Define collinearity and select methods for investigating it
- Interpret the variance-inflation factor and use it as a tool to choose variables
- Recall the procedures of the model selection approaches
- Execute and interpret the hypothesis testing and Information Theoretic (IT) approaches to model selection

GEORGE E P BOX

20TH CENTURY STATISTICIAN

All models are wrong, but some are useful

The Dilemma

- Measured many variables and want to include them all in a single modelling approach
- Interested in which covariates (and/or their interactions) are driving the response variable
  - E.g., you’re interested in changes of the number of saltmarsh sparrows and measured (12 explanatory variables):
    - % coverage of *Juncus gerardii*, Shrub, *Spartina patens*, *Distichlis spicata*, Bare ground, Other vegetation, *Phragmites australis,* Tall sedge, Water, *Spartina alterniflora* (short), *Spartina alterniflora* (tall)
    - Height of thatch, Max vegetation
    - Vegetation stem density
- What would you do?

The Steps

- Find the optimal set of covariates
- Steps:

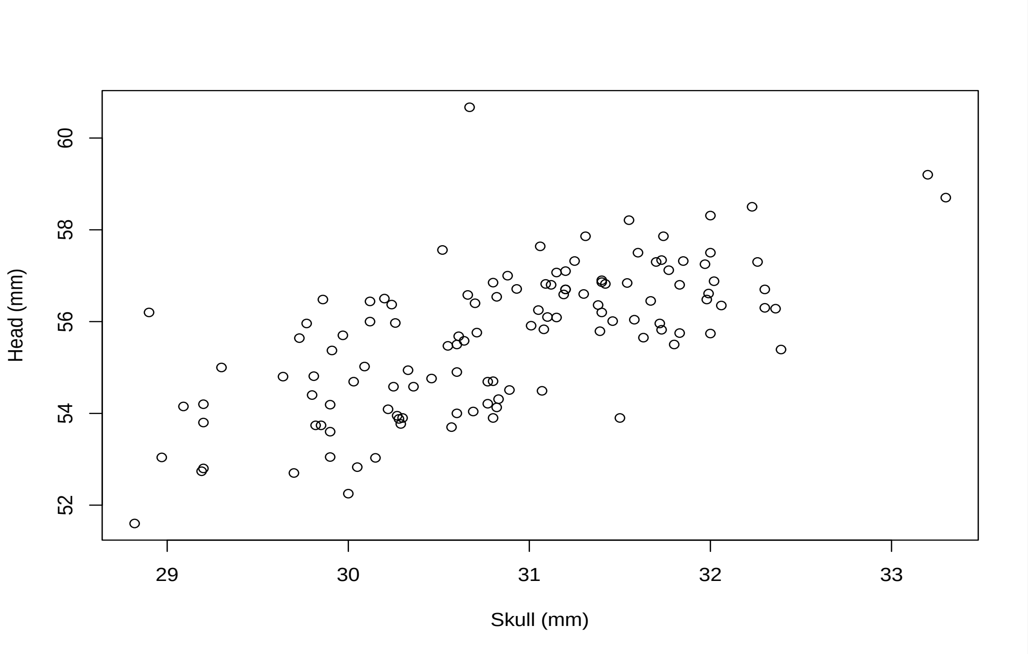
1. Check for redundancy (collinearity) amongst the explanatory variables and remove those variables
2. Fit your reduced model and decide on model selection procedures (if terms are non-significant)

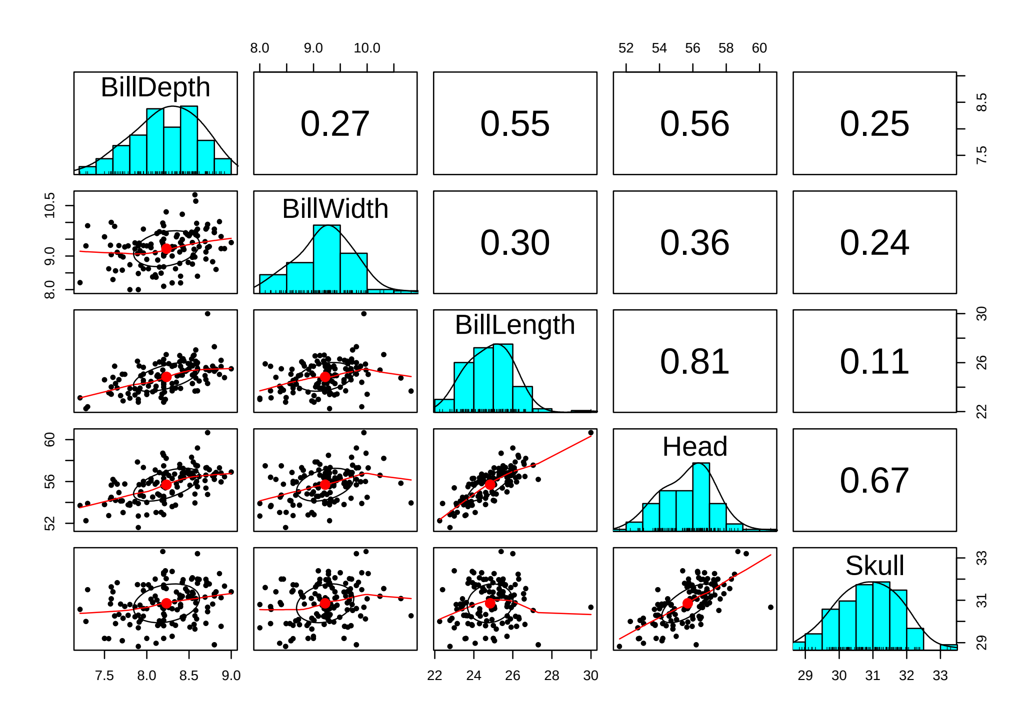
- Model selection is political and highly debated: some disciplines and journals advocate for certain methods
  - Debate among the staff at Imperial

Collinearity

Collinearity

- Correlation among our explanatory variables:
  - Body Mass ~ Head + Skull + Bill Depth + Bill Width + Bill Length
- If ignored increases Type II errors (false negatives)
  - Inflates standard errors via reducing in degrees of freedom (statistical power)
  - Confusing statistical analysis where nothing significant
- Methods to detect:
  - Plotting (pairwise plots amongst all variables)
  - Correlation tests
  - Variance-Inflation Factor (VIF)





Collinearity

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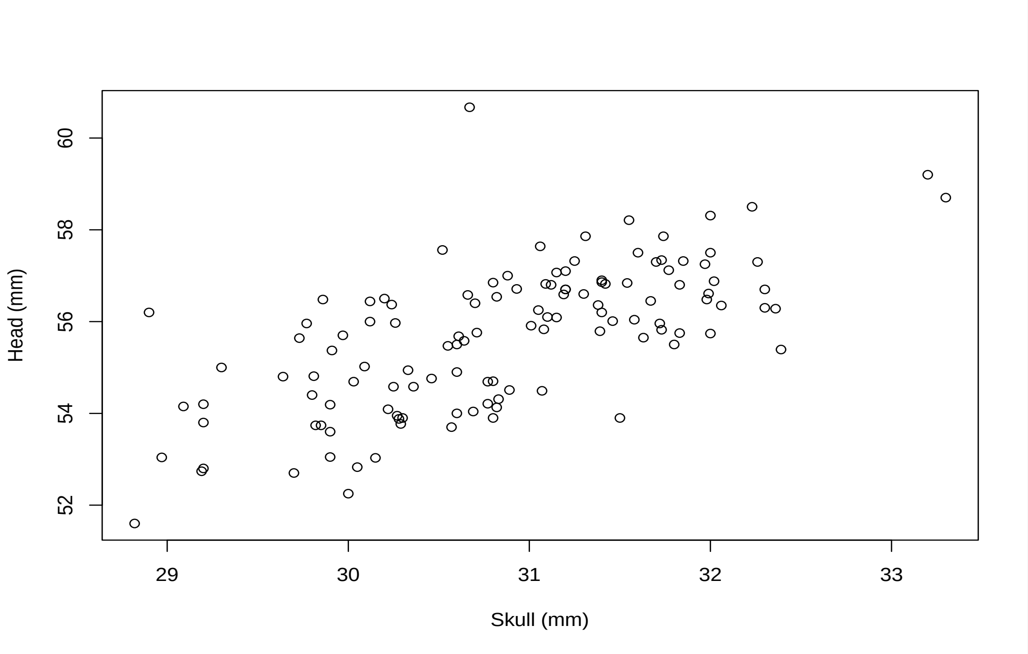
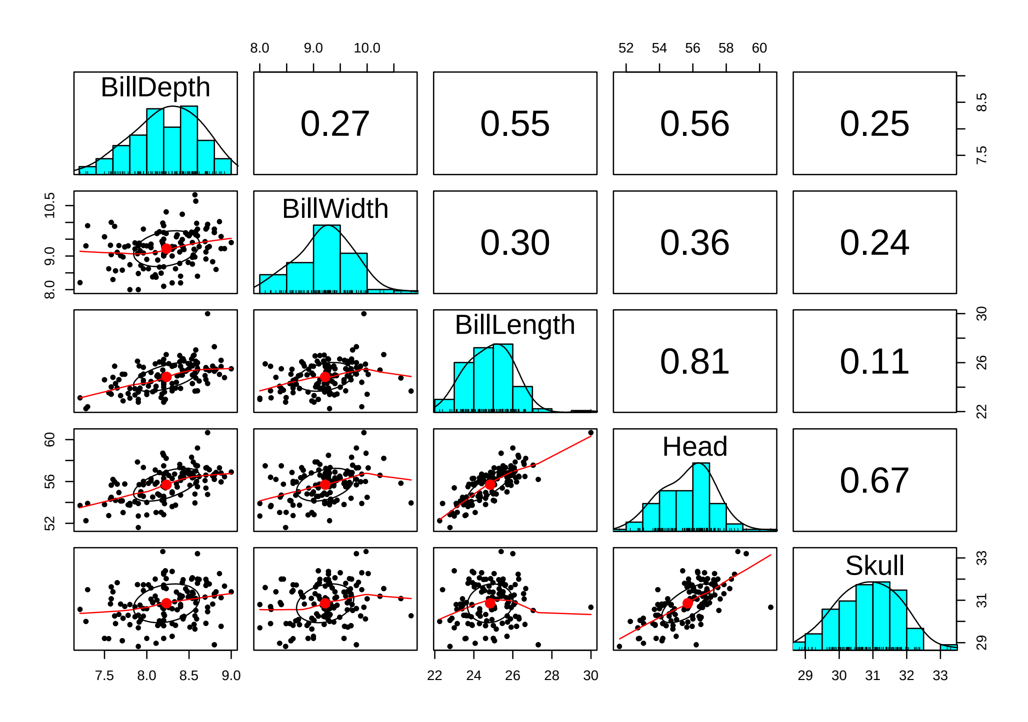
  - Plotting (pairwise plots amongst all variables)
  - Correlation tests
  - **Variance-Inflation Factor (VIF)**





Variance-Inflation Factor (VIF)

$$Variance(\beta_j) = \frac{1}{1 - R_j^2} * \frac{\sigma^2}{(n-1)S_j^2}$$

Variance of the residuals

Variance of coefficient of variable *j*

R^2 of model where covariate is fitted as response variable

Sample Size

Values of variable *j*

Variance-Inflation Factor (VIF)

Variance of the residuals

$$Variance(\beta_j) = \frac{1}{1 - R_j^2} * \frac{\sigma^2}{(n-1)S_j^2}$$

Variance of coefficient of variable *j*

Sample Size

Values of variable *j*

R^2 of model where covariate is fitted as response variable

Example 1

- Essentially, VIF is based on the R^2 of a series of models where each covariate becomes the response variable, e.g.
  - Body Mass ~ [Head + Skull + Bill Depth + Bill Width + Bill Length]
    - Head ~ Skull + Bill Depth + Bill Width + Bill Length
    - Skull ~ Head + Bill Depth + Bill Width + Bill Length
    - Bill Depth ~ Head + Skull + Bill Width + Bill Length
    - Bill Width ~ Head + Skull + Bill Depth + Bill Length
    - Bill Length ~ Head + Skull + Bill Depth + Bill Width
- Single VIF score for each variable

Variable	VIF
Head	2899993
Skull	98681.63
Bill Depth	1.54
Bill Width	1.17
Bill Length	162667.6

Extracted from R

What do we do with these numbers?

- Sequentially drop variable with the highest VIF, recalculate VIF and repeat until all variables are below a given threshold:
  - 10, 5 or 3 (depending on how conservative) → I'd go with 3.

Variable	VIF
Head	2899993
Skull	98681.63
Bill Depth	1.54
Bill Width	1.17
Bill Length	162667.6

→

Variable	VIF
Skull	1.11
Bill Depth	1.53
Bill Width	1.16
Bill Length	1.50

Example 2

- Number of Sparrows ~

Covariate	*P*-value (full model)	VIF	*P*-value (collinearity removed)
% *Juncus gerardii*	0·0203	44·9953	0·0001
% Shrub	0·9600	2·7818	0·0568
Height of thatch	0·9989	1·6712	0·8263
% *Spartina patens*	0·0640	159·3506	0·3312
% *Distichlis spicata*	0·0527	53·7545	0·2538
% Bare ground	0·0666	12·0586	0·8908
% Other vegetation	0·0730	5·8170	0·9462
% *Phragmites australis*	0·0715	3·7490	0·2734
% Tall sedge	0·2160	4·4093	0·4313
% Water	0·0568	17·0677	0·6942
% *Spartina alterniflora* (short)	0·0549	121·4637	0·2949
% *Spartina alterniflora* (tall)	0·0960	159·3828	
Maximum vegetation height	0·2432	6·1200	
Vegetation stem density	0·7219	3·2064	

Zuur, Ieno & Elphick (2010). *A protocol for data exploration to avoid common statistical problems*

What next? – Model Selection

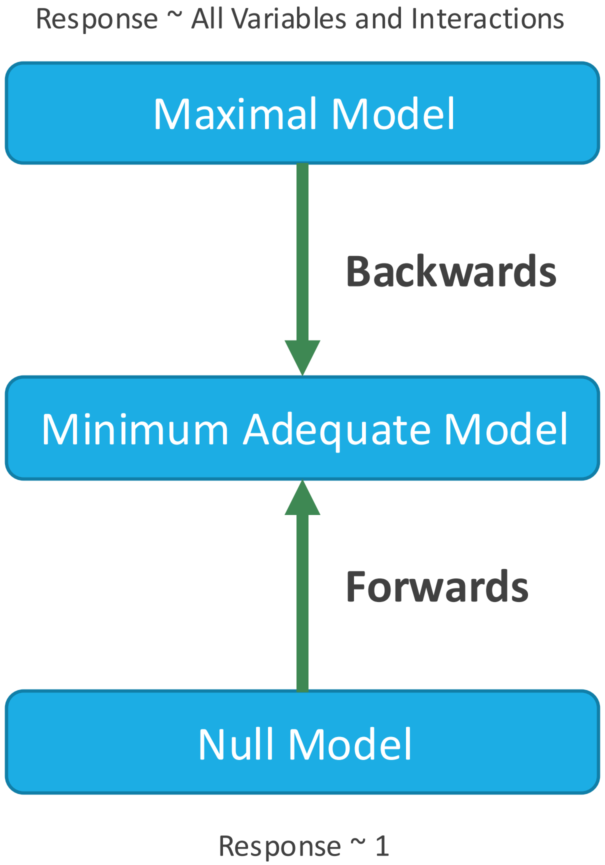
- After removing collinear variables, we can move onto model selection procedures

- Three approaches in ecology and evolution:
  1. *A priori* specification of a model and present the model from the outset (do nothing)
  2. Classic model selection using criteria
     - What approach? What criteria? What terms?
  3. *A priori* specification of a multiple models and comparison
     - Information Theoretic (IT) approach

Model Selection

Classic Model Selection – The Approach

- Finding the optimal model = *the minimum adequate model*
- Stepwise (forward or backward selection)
- Backwards is preferable due to *a prior* specification of the maximal
- Forwards, and to a certain extent backwards, is criticized for data dredging
- Backwards builds a maximal model and sequentially removes terms (higher order interactions first and then main effects)
- Debated whether only interactions should be removed
- Removal based on hypothesis testing or information criteria

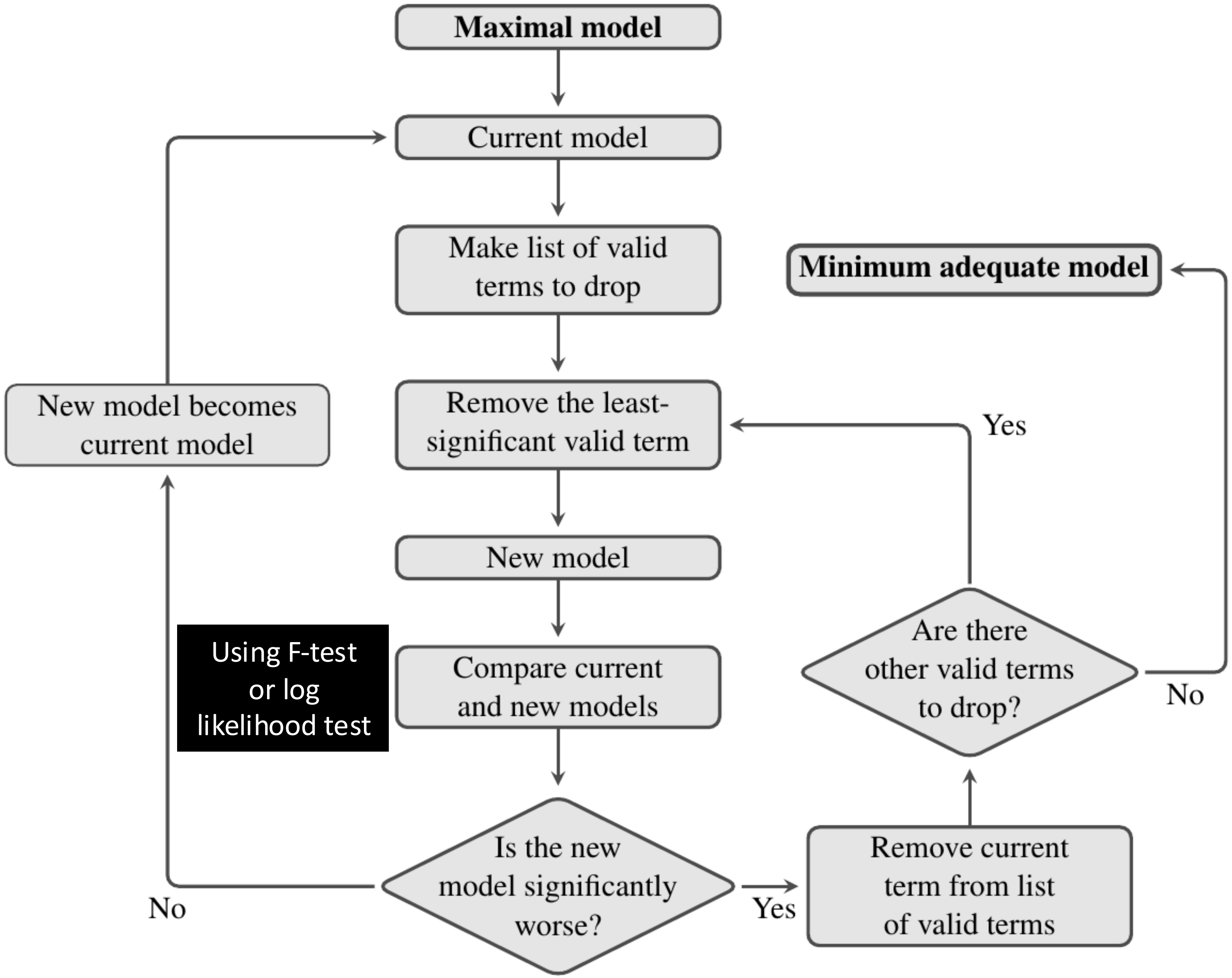


Classic Model Selection – The Criteria

1. Hypothesis testing procedure
   - Removing terms hierarchically based on highest p-values
   - Starting with higher order interactions to main effects
   - Removing terms will reduce the R^2 (explanatory power) of your model and you evaluate whether this removal is negligible or not.
2. Information criteria
   - Estimators of prediction error = quality of the model
     - Akaike Information Criterion (AIC)
     - Bayesian Information Criterion (BIC)

Hypothesis Testing Procedure

- Rules:
  1. Can’t remove a significant term
  2. Can’t remove a main effect if part of an interaction
  3. Always performed iteratively (removing one term at a time)



TheMulQuaBio: Dr Samraat Pawar

Example

```
Call:
lm(formula = log(BMR..W.) ~ log(Mass..g.) * Food * Climate, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.86449 -0.24211 -0.00055  0.19882  0.95438 

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)    
(Intercept)                   -3.87740    0.13229 -29.310  < 2e-16 ***
log(Mass..g.)                  0.72847    0.01960  37.165  < 2e-16 ***
FoodI                          0.43199    0.17374   2.486   0.0141 *  
ClimateTR                     -1.11265    0.70226  -1.584   0.1154    
log(Mass..g.):FoodI           -0.15781    0.03427  -4.605 9.25e-06 ***
log(Mass..g.):ClimateTR        0.12950    0.07351   1.762   0.0803 .  
FoodI:ClimateTR                0.55724    0.71932   0.775   0.4399    
log(Mass..g.):FoodI:ClimateTR -0.01837    0.08316  -0.221   0.8255    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3669 on 138 degrees of freedom
  (6 observations deleted due to missingness)
Multiple R-squared:  0.9692,	Adjusted R-squared:  0.9676 
F-statistic:   620 on 7 and 138 DF,  p-value: < 2.2e-16
```

```
Analysis of Variance Table

Response: log(BMR..W.)
                            Df Sum Sq Mean Sq  F value    Pr(>F)    
log(Mass..g.)                1 576.75  576.75 4285.0054 < 2.2e-16 ***
Food                         1   2.75    2.75   20.4488 1.307e-05 ***
Climate                      1   0.28    0.28    2.0470  0.154770    
log(Mass..g.):Food           1   2.60    2.60   19.3475 2.162e-05 ***
log(Mass..g.):Climate        1   1.44    1.44   10.7063  0.001349 ** 
Food:Climate                 1   0.29    0.29    2.1493  0.144909    
log(Mass..g.):Food:Climate   1   0.01    0.01    0.0488  0.825485    
Residuals                  138  18.57    0.13                       
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Analysis of Variance Table

Model 1: log(BMR..W.) ~ log(Mass..g.) * Food * Climate
Model 2: log(BMR..W.) ~ log(Mass..g.) + Food + Climate + log(Mass..g.):Food + 
    log(Mass..g.):Climate + Food:Climate
  Res.Df    RSS Df  Sum of Sq      F Pr(>F)
1    138 18.574                            
2    139 18.581 -1 -0.0065688 0.0488 0.8255
```

Example 2

	14		11	2
Covariate	*P*-value (full model)	VIF	*P*-value (collinearity removed)	*P*-value (reduced model)
% *Juncus gerardii*	0·0203	44·9953	0·0001	0·00004
% Shrub	0·9600	2·7818	0·0568	0·0727
Height of thatch	0·9989	1·0712	0·8265	
% *Spartina patens*	0·0640	159·3506	0·3312	
% *Distichlis spicata*	0·0527	53·7545	0·2538	
% Bare ground	0·0666	12·0586	0·8908	
% Other vegetation	0·0730	5·8170	0·9462	
% *Phragmites australis*	0·0715	3·7490	0·2734	
% Tall sedge	0·2160	4·4093	0·4313	
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% *Spartina alterniflora* (short)	0·0549	121·4637	0·2949	
% *Spartina alterniflora* (tall)	0·0960	159·3828		
Maximum vegetation height	0·2432	6·1200		
Vegetation stem density	0·7219	3·2064		

Mixed Model Selection

Mixed Effects Model Selection

1. Specify the fixed effects for maximal model and fit using the gls() function.
2. Specify the random effect structure and fit using lmer() adding the argument REML=TRUE.
3. Compare gls() and lmer() model using log-likelihood test
4. Iteratively remove the random effects and test sequentially using the log-likelihood test
5. Iteratively remove the fixed effects and test sequentially using the log-likelihood test
6. Refit the minimum adequate model with REML=FALSE, validate the model and interpret the coefficients.
7. Present the results in a table and make a supporting figure

Example – Owl Data

- Sampled different 599 nestlings from 27 nests
- Measured:
  - Number of calls to a visit from a parent per nestling
  - The sex of the visiting parent
  - How long it took the parent to arrive
  - Whether the owls were in a deprived or satiated treatment
  - The nest id
- Research question:
  - What factors influence the number of calls of nestlings?

Steps 1-3

```
M1<- gls(NegPerChick~FoodTreatment*SexParent*ArrivalTime, data=Owls)
M2<- lmer(NegPerChick~FoodTreatment*SexParent*ArrivalTime+(1|Nest), data=Owls, REML = TRUE)
lrtest(M1, M2)
```

```
Likelihood ratio test

Model 1: NegPerChick ~ FoodTreatment * SexParent * ArrivalTime
Model 2: NegPerChick ~ FoodTreatment * SexParent * ArrivalTime + (1 |
    Nest)
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   9 -1108.4
2  10 -1100.0  1 16.902  3.935e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Log-likelihood has increased by 8.4 = model accuracy/quality has significantly improved

Step 4

```
Linear mixed model fit by REML ['lmerMod']
Formula: NegPerChick ~ FoodTreatment * SexParent * ArrivalTime + (1 |      Nest)
   Data: Owls

REML criterion at convergence: 2200

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9010 -0.6884 -0.2621  0.5146  4.9993

Random effects:
 Groups   Name        Variance Std.Dev.
 Nest     (Intercept) 0.2301   0.4797
 Residual             2.1344   1.4610
Number of obs: 599, groups:  Nest, 27
```

Nest variance > 0 and accounts for non-independence so should keep

Step 5

```
Fixed effects:
                                        Estimate Std. Error t value
(Intercept)                              7.51039    1.69859   4.422
FoodTreatmentSatiated                   -2.46045    2.42787  -1.013
SexParentMale                            1.23164    2.16643   0.569
ArrivalTime                             -0.22294    0.06827  -3.265
FoodTreatmentSatiated:SexParentMale     -2.68301    3.22559  -0.832
FoodTreatmentSatiated:ArrivalTime        0.06219    0.09792   0.635
SexParentMale:ArrivalTime               -0.04835    0.08699  -0.556
FoodTreatmentSatiated:SexParentMale:ArrivalTime  0.11279    0.13006   0.867
```

Non-significant highest order interaction

Step 4

```
Linear mixed model fit by REML ['lmerMod']
Formula: NegPerChick ~ FoodTreatment * SexParent * ArrivalTime + (1 |      Nest)
   Data: Owls

REML criterion at convergence: 2200

Scaled residuals:
    Min      1Q  Median      3Q     Max 
-1.9010 -0.6884 -0.2621  0.5146  4.9993 

Random effects:
 Groups   Name        Variance Std.Dev.
 Nest     (Intercept) 0.2301   0.4797  
 Residual             2.1344   1.4610  
Number of obs: 599, groups:  Nest, 27

Fixed effects:
```

	Estimate	St	
(Intercept)	7.51039		
FoodTreatmentSatiated	-2.46045		
SexParentMale	1.23164		
ArrivalTime	-0.22294		
FoodTreatmentSatiated:SexParentMale	-2.68301		
FoodTreatmentSatiated:ArrivalTime	0.06219	0.09792	0.635
SexParentMale:ArrivalTime	-0.04835	0.08699	-0.556
FoodTreatmentSatiated:SexParentMale:ArrivalTime	0.11279	0.13006	0.867

Step 5

Nest variance > 0 and accounts for non-independence so should keep

Analysis of Variance Table

	npar	Sum Sq	Mean Sq	F value
FoodTreatment	1	93.993	93.993	44.0375
SexParent	1	0.241	0.241	0.1130
ArrivalTime	1	80.782	80.782	37.8480
FoodTreatment:SexParent	1	0.447	0.447	0.2096
FoodTreatment:ArrivalTime	1	8.161	8.161	3.8237
SexParent:ArrivalTime	1	0.003	0.003	0.0014
FoodTreatment:SexParent:ArrivalTime	1	1.605	1.605	0.7521

Non-significant highest order interaction

Step 5

```
M3<- update(M2,.~.-FoodTreatment:SexParent:ArrivalTime)
lrtest(M2, M3)
```

```
Likelihood ratio test

Model 1: NegPerChick ~ FoodTreatment * SexParent * ArrivalTime + (1 |
    Nest)
Model 2: NegPerChick ~ FoodTreatment + SexParent + ArrivalTime + (1 |
    Nest) + FoodTreatment:SexParent + FoodTreatment:ArrivalTime +
    SexParent:ArrivalTime
  #Df  LogLik Df  Chisq Pr(>Chisq)
1  10 -1100.0
2   9 -1099.2 -1 1.4928     0.2218
```

Log-likelihood has increased by 0.8, no differences between the models but M3 is simpler than M2, repeat step 5

Step 6

```
M7 <- update(M6, REML=FALSE)
summary(M7)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: NegPerChick ~ FoodTreatment + SexParent + ArrivalTime + (1 |      Nest)
   Data: Owls

     AIC      BIC   logLik deviance df.resid
  2194.2   2220.6  -1091.1   2182.2      593

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.8537 -0.6805 -0.2237  0.5265  4.9483
```

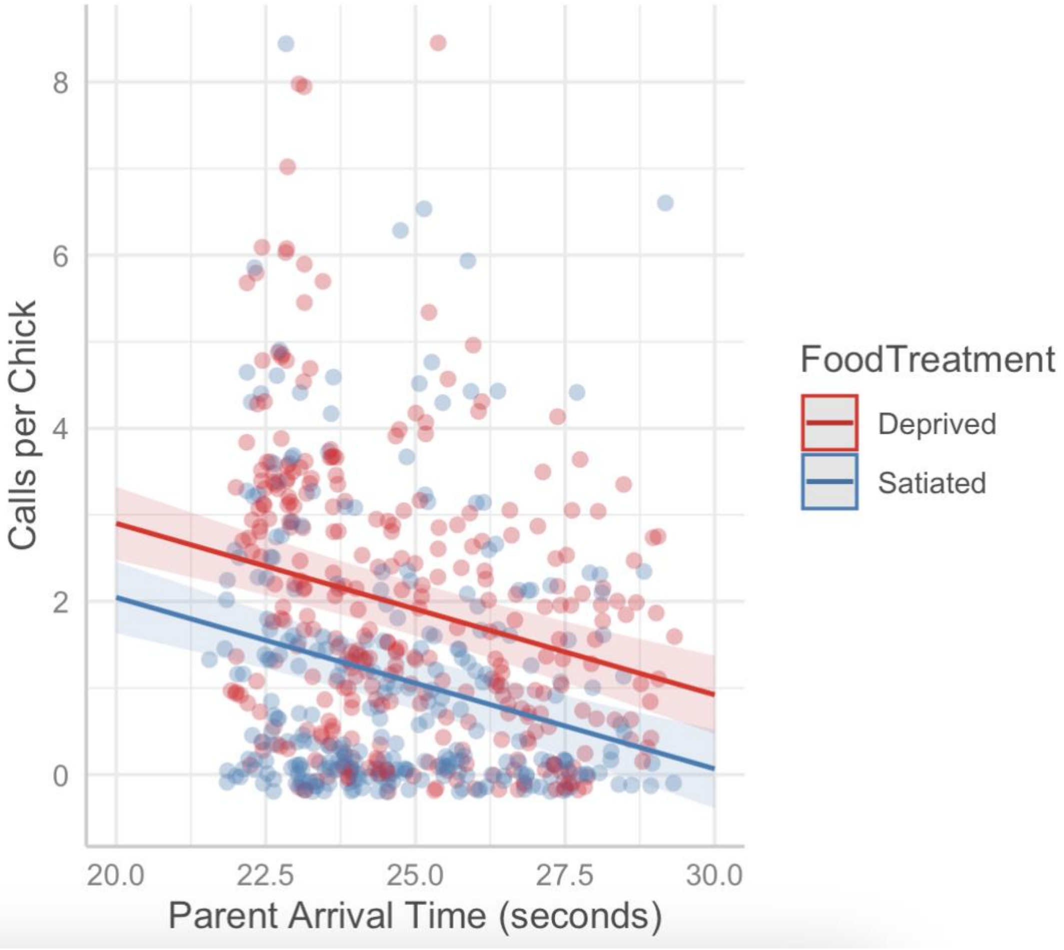
```
Random effects:
 Groups   Name        Variance Std.Dev.
 Nest     (Intercept) 0.2392   0.4891
 Residual             2.1200   1.4560
Number of obs: 599, groups:  Nest, 27
```

Nest explains 10.14% of variation in calls per chick

```
Fixed effects:
                      Estimate Std. Error t value
(Intercept)            6.86017    0.80656   8.505
FoodTreatmentSatiated -0.85527    0.12513  -6.835
SexParentMale          0.08250    0.13259   0.622
ArrivalTime           -0.19795    0.03206  -6.173
```

Chicks had fewer calls when satiated and when their parents took longer to arrive. A one second increase in arrival time decreased the number of calls per chick by -0.20±0.03 (t=-6.17). Sex of the parent had no effect on chick calls (0.08±0.13, t=0.62).

Step 7



	Calls per Chick		
Predictors	*Estimates*	*CI*	*p*
(Intercept)	6.86	5.28 – 8.44	**<0.001**
FoodTreatment [Satiated]	-0.86	-1.10 – -0.61	**<0.001**
SexParent [Male]	0.08	-0.18 – 0.34	0.534
ArrivalTime	-0.20	-0.26 – -0.13	**<0.001**
Random Effects			
σ^2	2.12		
$\tau_{00\ Nest}$	0.24		
ICC	0.10		
N_{Nest}	27		
Observations	599		
Marginal R^2 / Conditional R^2	0.118 / 0.207		

```
require(ggeffects)
M7_plot <- ggpredict(M7, terms=c("ArrivalTime","FoodTreatment"))
plot(M7_plot, add.data = T, jitter = TRUE)+
  labs(y="Calls per Chick", x="Parent Arrival Time (seconds)", title="")
```

```
require(sjPlot)
tab_model(M7,dv.labels = "Calls per Chick")
```

sjPlot – tab_model()

	Calls per Chick		
Predictors	*Estimates*	*CI*	*p*
(Intercept)	6.86	5.28 – 8.44	**<0.001**
FoodTreatment [Satiated]	-0.86	-1.10 – -0.61	**<0.001**
SexParent [Male]	0.08	-0.18 – 0.34	0.534
ArrivalTime	-0.20	-0.26 – -0.13	**<0.001**
Random Effects			
σ^2	2.12	Residual Variance	
τ_{00} Nest	0.24	Nest Variance	
ICC	0.10	Proportion explained	
N Nest	27		
Observations	599		
Marginal R^2 / Conditional R^2	0.118 / 0.207		

	Maximal Model			Minimum Adequate Model		
Predictors	*Estimates*	*CI*	*p*	*Estimates*	*CI*	*p*
(Intercept)	7.51	4.20 – 10.83	**<0.001**	6.86	5.28 – 8.44	**<0.001**
FoodTreatment [Satiated]	-2.47	-7.21 – 2.27	0.307	-0.86	-1.10 – -0.61	**<0.001**
SexParent [Male]	1.23	-3.00 – 5.46	0.567	0.08	-0.18 – 0.34	0.534
ArrivalTime	-0.22	-0.36 – -0.09	**0.001**	-0.20	-0.26 – -0.13	**<0.001**
FoodTreatment [Satiated] × SexParent [Male]	-2.71	-9.01 – 3.58	0.397			
FoodTreatment [Satiated] × ArrivalTime	0.06	-0.13 – 0.25	0.521			
SexParent [Male] × ArrivalTime	-0.05	-0.22 – 0.12	0.577			
(FoodTreatment [Satiated] × SexParent [Male]) × ArrivalTime	0.11	-0.14 – 0.37	0.378			
Random Effects						
σ^2	2.11			2.12		
τ_{00}	0.21 Nest			0.24 Nest		
ICC	0.09			0.10		
N	27 Nest			27 Nest		
Observations	599			599		
Marginal R^2 / Conditional R^2	0.125 / 0.205			0.118 / 0.207		

```
tab_model(M2, M7, dv.labels=c("Maximal Model", "Minimum Adequate Model"))
```

Information Criteria

MODEL SELECTION

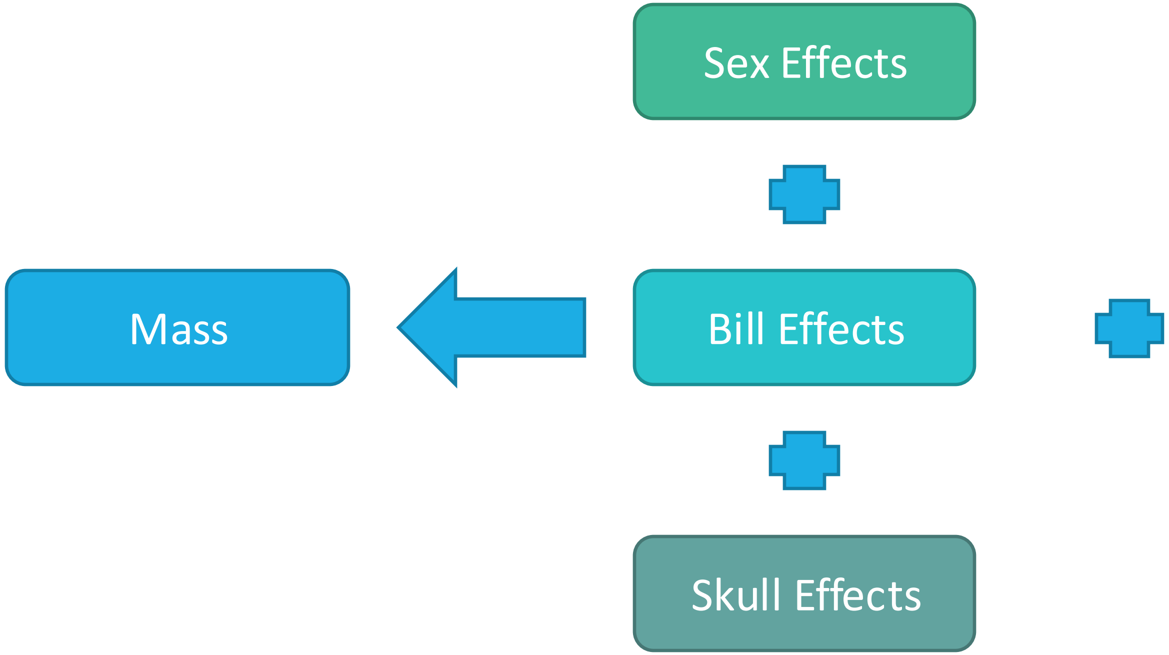
Information Criteria – AIC

$$AIC = -2 * logLik + 2 * p$$

- Combines model fit via log-likelihood and model complexity (p= number of parameters estimated).
- AIC’s are used to compared models built on the same response variable and dataset
- Preferable to avoid using AIC on models with small sample sizes (<50)
  - Can account for this by using AICc instead
- Choose the model with the lowest AIC value
- `step` function automates the selection procedure (see practical later)
- AIC is integral in the Information Theoretic (IT) Approach

The Information Theoretic (IT) Approach

- Gaining popularity in ecological fields and dictates that we thoroughly assess the data with a range of plausible ecological hypotheses
- Using *a prior* knowledge construct 10-15 models
- Models are compared with AIC and the most interesting/important range of models are presented
- Different to hypothesis testing as it produces a range of models rather than just one



The Information Theoretic (IT) Approach

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Model	Description
Skull + Bill Length + Bill Depth + Bill Width + Sex	All variables (additive)
Skull + Bill Length + Bill Depth + Bill Width + Sex + Sex:Skull + Sex:Bill Length + Sex:Bill Depth + Sex:Bill Width	All variables (interactions with sex)
Skull + Bill Length + Bill Depth + Bill Width	All variables (no sex)
Skull + Sex	Skull and sex
Skull + Sex + Sex:Skull	Skull interacting with sex
Skull	Just Skull
Bill Length + Bill Depth + Bill Width + Sex	Bill traits and sex
Bill Length + Bill Depth + Bill Width + Sex + Sex:Bill Length + Sex:Bill Depth + Sex:Bill Width	Bill traits interaction with sex
Bill Length + Bill Depth + Bill Width	Just Bill traits

The Information Theoretic (IT) Approach

Model	Description	AIC	AIC Differences	Akaike Weights
Skull + Bill Length + Bill Depth + Bill Width + Sex	All variables (additive)	665.70	0	0.65
Skull + Bill Length + Bill Depth + Bill Width + Sex + Sex:Skull + Sex:Bill Length + Sex:Bill Depth + Sex:Bill Width	All variables (interactions with sex)	671.41	5.71	0.04
Skull + Bill Length + Bill Depth + Bill Width	All variables (no sex)	667.17	1.47	0.31
Skull + Sex	Skull and sex	691.82	26.12	0
Skull + Sex + Sex:Skull	Skull interacting with sex	693.79	28.09	0
Skull	Just Skull	693.35	27.65	0
Bill Length + Bill Depth + Bill Width + Sex	Bill traits and sex	704.11	38.41	0
Bill Length + Bill Depth + Bill Width + Sex + Sex:Bill Length + Sex:Bill Depth + Sex:Bill Width	Bill traits interaction with sex	707.86	42.16	0
Bill Length + Bill Depth + Bill Width	Just Bill traits	702.76	37.06	0
Sex	Just sex	721.15	55.44	0

The Information Theoretic (IT) Approach

- Models <2 AICs from best model are indistinguishable
- From the AICs, we can calculate Akaike Weights
- The Akaike Weights represent the relative likelihood of that model
- If we were to repeat the experiment a large number of times, in 65% of cases the model with all variable (additive) would be the optimal model, and in 31% of cases the all variable but no sex would the optimal model.
- For the IT approach, we'd present and interpret both models **OR**

Model	Description	AIC	AIC Diff	Akaike Weights
Skull + Bill Length + Bill Depth + Bill Width + Sex	All variables (additive)	665.70	0	0.65
Skull + Bill Length + Bill Depth + Bill Width + Sex + Sex:Skull + Sex:Bill Length + Sex:Bill Depth + Sex:Bill Width	All variables (interactions with sex)	671.41	5.71	0.04
Skull + Bill Length + Bill Depth + Bill Width	All variables (no sex)	667.17	1.47	0.31

Model Averaging

- Averaging of the coefficients of models <2 AICs from the best model (lowest AIC).
- Two approaches:
  - Conditional averaging – average coefficients in which the variable exists

	Skull	Bill Length	Bill Width	Bill Depth	Sex
Model1	0.25	1.34	1.44	1.56	0.5
Model2	0.19	1.34	1.47	1.86	--
Average	0.22	1.34	1.46	1.71	0.5

Model Averaging

- Averaging of the coefficients of models <2 AICs from the best model (lowest AIC).
- Two approaches:
  - Conditional averaging – average coefficients in which the variable exists
  - Full averaging – average coefficients whereby missed selection receive 0

	Skull	Bill Length	Bill Width	Bill Depth	Sex
Model1	0.25	1.34	1.44	1.56	0.5
Model2	0.19	1.34	1.47	1.86	0
Average	0.22	1.34	1.46	1.71	0.25

- This discourse reiterates the message of:

GEORGE E P BOX

20TH CENTURY STATISTICIAN

All models are wrong, but some are useful

Summary

- Investigating and adjusting statistical models according to collinearity is vital to reduce Type II errors
- The variance-inflation factor can be used to whittle down the number of variables in a model but the threshold for this is debatable (3, 5 or 10)
- Model selection is a controversial topic in statistics and ecological modelling and various approaches exists; from do nothing to data dredging approaches like stepwise forward selection
- Stepwise backwards selection involved removing terms sequentially from a model evaluating the reduced model to the previous model in finding the minimum adequate model
- The Information Theoretic (IT) approach produces a range of candidate models based on prior knowledge, of which all should be interpreted and presented